The Probability Scale and Single Events

Name _____ Date _____ No calculator Version 1 Scan to practise this topic online

KEY WORDS probability *(tūponotanga)* · outcome *(putanga)* · event *(takahanga)* · equally likely *(tūpono ōrite)*

A The probability scale ●○○

Remember: A probability is a number from 0 (impossible) to 1 (certain). An even chance is $\frac{1}{2}$.

1. Choose the best description of each event: impossible, unlikely, even chance, likely or certain.

a) Rolling a 9 on an ordinary dice _____

b) Getting a tail when a coin is tossed _____

c) Rolling a number less than 10 on a dice _____

d) Rolling a 3 on a dice _____

e) Picking a red counter from a bag of 7 red and 1 blue _____

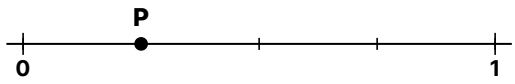
f) Rolling an odd number on a dice _____

g) Picking a blue counter from a bag of 2 blue and 8 red _____

h) Rolling a number greater than 1 on a dice _____

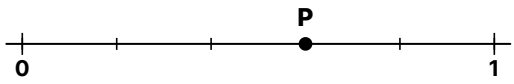
2. Each number line goes from 0 to 1. Write the probability shown by the point P as a fraction.

a)



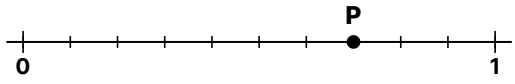
$P =$ ____

b)



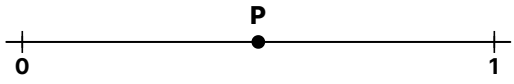
$P =$ ____

c)



$P =$ ____

d)



$P =$ ____

B Probabilities of single events ●●○

Remember: Probability of an event = the number of ways it can happen ÷ the total number of outcomes. Give fractions in their simplest form.

3. An ordinary dice is rolled. Find the probability of rolling:

a) a 4 **b)** an odd number **c)** a number less than 3

d) a 1 or a 6 **e)** a number greater than 6 **f)** a number from 1 to 6

4. A bag holds 4 red, 3 blue and 1 green counters. One counter is taken out without looking. Find the probability that it is:

a) red **b)** blue **c)** green **d)** not blue

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5. A spinner has 8 equal sections numbered 1 to 8. Find the probability that it lands on:

a) 5 **b)** an even number **c)** a number greater than 5 **d)** a prime number

6. Cards numbered 1 to 10 are shuffled and one card is picked. Find the probability that the number is:

a) even **b)** a multiple of 3 **c)** less than 4 **d)** a two-digit number

C An event not happening ●●○

> **Remember:** An event either happens or it does not, so the two probabilities add to 1. With percentages they add to 100%.

7. The probability that an event happens is given. Find the probability that it does not happen.

a) 0.2 **b)** $\frac{1}{4}$ **c)** 30%

d) 0.65 **e)** $\frac{3}{8}$ **f)** 85%

g) $\frac{5}{6}$ **h)** 0.07 **i)** 1%

D Problems ●●●

8. A box holds 20 pens. The probability of picking a red pen is $\frac{1}{4}$. How many red pens are in the box?

9. A football match can be won, drawn or lost. The probability of a win is 0.3 and the probability of a draw is 0.45. Find the probability of a loss.

10. A class has 12 girls and 9 boys. One student is chosen at random to answer a question. Find the probability that a girl is chosen.

Extension

11. A bag holds 3 red counters, 5 blue counters and some yellow counters. The probability of picking a yellow counter is $\frac{1}{3}$. How many yellow counters are there?

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The Probability Scale and Single Events

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KEY WORDS probability *(tūponotanga)* · outcome *(putanga)* · event *(takahanga)* · equally likely *(tūpono ōrite)*

A The probability scale ●○○

Remember: A probability is a number from 0 (impossible) to 1 (certain). An even chance is $\frac{1}{2}$.

1. Choose the best description of each event: impossible, unlikely, even chance, likely or certain.

a) Rolling a 0 on an ordinary dice _____

b) Getting a head when a coin is tossed _____

c) Rolling a number less than 8 on a dice _____

d) Rolling a 5 on a dice _____

e) Picking a green counter from a bag of 9 green and 1 red _____

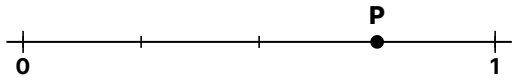
f) Rolling an even number on a dice _____

g) Picking a red counter from a bag of 1 red and 7 blue _____

h) Rolling a number less than 6 on a dice _____

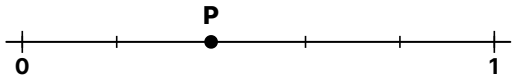
2. Each number line goes from 0 to 1. Write the probability shown by the point P as a fraction.

a)



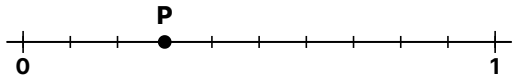
$P =$ ____

b)



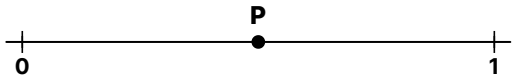
$P =$ ____

c)



$P =$ ____

d)



$P =$ ____

B Probabilities of single events ●●○

Remember: Probability of an event = the number of ways it can happen ÷ the total number of outcomes. Give fractions in their simplest form.

3. An ordinary dice is rolled. Find the probability of rolling:

a) a 6 **b)** an even number **c)** a number greater than 4

d) a 2 or a 3 **e)** a 7 **f)** a number less than 7

4. A bag holds 3 red, 5 blue and 2 green counters. One counter is taken out without looking. Find the probability that it is:

a) red **b)** blue **c)** green **d)** not blue

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5. A spinner has 8 equal sections numbered 1 to 8. Find the probability that it lands on:

a) 5 **b)** an even number **c)** a number greater than 5 **d)** a prime number

6. Cards numbered 1 to 12 are shuffled and one card is picked. Find the probability that the number is:

a) even **b)** a multiple of 3 **c)** less than 4 **d)** a two-digit number

C An event not happening ●●○

> **Remember:** An event either happens or it does not, so the two probabilities add to 1. With percentages they add to 100%.

7. The probability that an event happens is given. Find the probability that it does not happen.

a) 0.6 **b)** $\frac{1}{5}$ **c)** 40%

d) 0.35 **e)** $\frac{5}{8}$ **f)** 75%

g) $\frac{1}{6}$ **h)** 0.09 **i)** 2%

D Problems ●●●

8. A box holds 24 pens. The probability of picking a red pen is $\frac{1}{3}$. How many red pens are in the box?

9. A football match can be won, drawn or lost. The probability of a win is 0.2 and the probability of a draw is 0.55. Find the probability of a loss.

10. A class has 15 girls and 11 boys. One student is chosen at random to answer a question. Find the probability that a girl is chosen.

Extension

11. A bag holds 4 red counters, 6 blue counters and some yellow counters. The probability of picking a yellow counter is $\frac{1}{3}$. How many yellow counters are there?

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